the same honor. Kahneman and Tversky formulated prospect theory based on how people actually make decisions (it was a positive theory).[20]

There are two parts to prospect theory:

1. Loss aversion utility

   Investors find the pain of losses to be greater than the joy from gains. Loss aversion utility allows the investor to have different slopes (marginal utilities) for gains and losses.
2. Probability transformations

   Kahneman and Tversky move beyond probabilities—even subjective probabilities. They transform probabilities to *decision weights*, which do not necessarily obey probability laws. They do not, for example, have to sum to one like probabilities. Decision weights allow investors to potentially severely overweight low probability events—including both disasters and winning the lottery.

We can write prospect theory as a form of expected utility similar to equation (2.1):

$$U = \sum_s w(p_s)U(W_s), \tag{2.11}$$

for a probability weight function $w(\bullet)$ and a loss aversion function $U(\bullet)$, which operate over different states s, which occur with probability p_s.

Let's focus on loss aversion utility. Figure 2.10 graphs a typical loss aversion utility function. Utility is defined relative to a reference point, which is the origin on the graph. The reference point can be zero (absolute return), or it could be a risk-free rate (sure return) or a risky asset return (benchmark return). Gains are defined as positive values on the x-axis relative to the origin, and losses are defined as negative values. The utility function has a kink at the origin, so there is asymmetry in how investors treat gains and losses. The utility function over gains is concave, so investors are risk averse over gains. The shape of the utility function for losses is *convex*, which captures the fact that people are *risk seeking over losses*—that is, they are willing to accept some risk to avert a certain loss. The utility function for losses is also steeper than the utility function over gains. Thus, investors are more sensitive to losses than to gains.

The fact that investors are risk seeking over losses is borne out in many experiments that involve choosing between bets like:

A: Losing $1,000 with probability of 0.5.
B: Losing $500 for sure.

[20] For a summary of applications of prospect theory and other behavioral theories to finance, see Barberis and Thaler (2003).